StadiumAI

AI-powered crowd safety and volunteer coordination for live stadium events — turning chaotic field updates into coordinated, AI-guided action in real time.

HACKATHON DEMO

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The Problem: Seconds Matter

During large events, crowd incidents escalate in seconds — but traditional response systems simply cannot keep up.

📨 Unstructured Updates

Volunteers send freeform messages with no standard format, making triage slow and unreliable.

🔀 Signal Overload

Control rooms struggle to validate and prioritise incoming reports under pressure.

📡 Delayed Broadcasts

Critical alerts reach responders too late, widening the window of risk for attendees.

⊗ **Delayed response = higher risk.** Every extra second without a coordinated action increases crowd danger.

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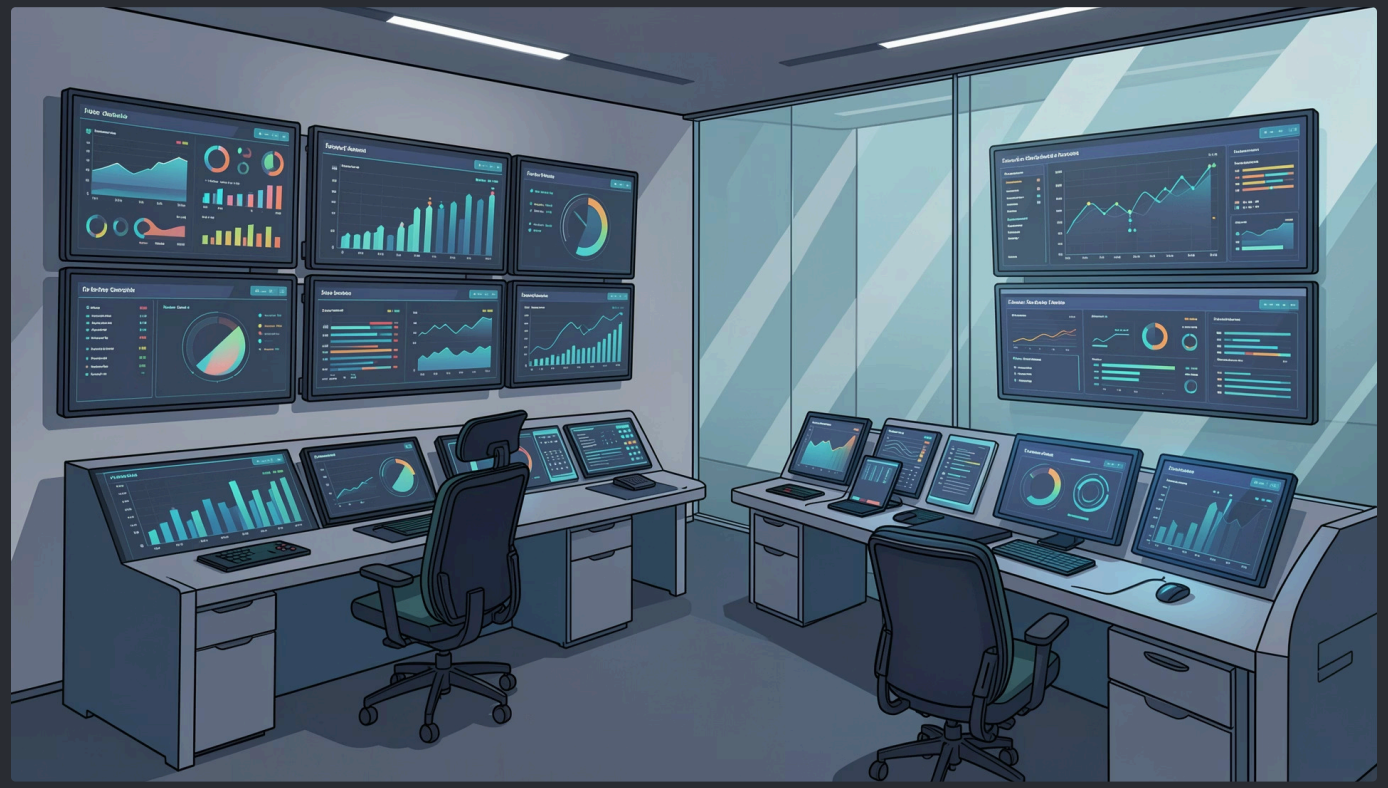
Introducing StadiumAI



A real-time AI operations system built for the field

- 🧠 AI analyses volunteer text and photo inputs instantly
- 📊 Dashboard reflects live crowd conditions by zone
- 📡 Telegram bot enables fast, structured field communication
- 🚨 Automated alerts with AI-generated action recommendations

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Core Features

Command Dashboard

Live zone heatmap with crowd density percentages, emergency banners, AI-suggested actions, and volunteer activity logs — all in one view.

Telegram Workflow

Guided onboarding, persistent action menus, instant status updates, photo submissions, and auto-broadcast to all active volunteers.

AI Incident Intelligence

Classifies crowd density and emergency type from text or images. Outputs structured density, emergency flag, summary, and suggested next action.

Emergency Orchestration

Any emergency report is instantly tagged, escalated, and fan-out broadcast to all registered volunteers — with AI suggestions visible on the admin view.

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Volunteer Workflow: Telegram Bot

What volunteers can do in the field

01

Register — Enter volunteer ID and zone assignment via `/start`

02

Update — Send text or photo status reports any time

03

Alert — Trigger emergency broadcast mode instantly

04

Monitor — View live crowd status and AI insights on demand

05

Sign Out — Close shift cleanly when done

Live bot: **@StadiumAI_aplbot** on Telegram — try it now during the demo.

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Emergency in Action: A Real Scenario

Report
Volunteer sends "Gate A fire and smoke"

Detect
AI identifies emergency intent and crowd risk

Update Zone
Zone shows emergency message and AI suggestion

Dashboard
Dashboard switches to emergency state

Broadcast
All volunteers receive instant alert

From a single field message to a fully coordinated response in seconds. AI suggests: **"Divert crowd to Gate B and dispatch fire response."** — no manual triage required.

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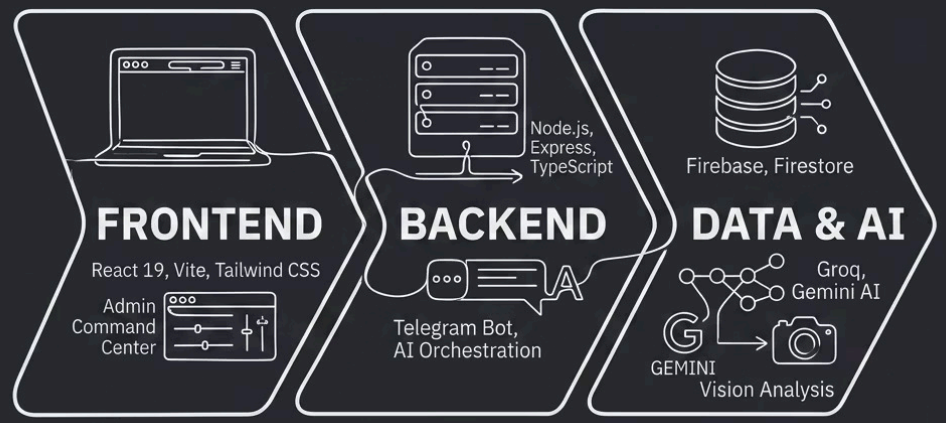
System Architecture



Built to scale under pressure

- **Frontend:** React 19, Vite, Tailwind CSS, Framer Motion
- **Backend:** Node.js, Express, TypeScript (tsx runtime)
- **Bot:** node-telegram-bot-api
- **Database:** Firebase Firestore + Admin SDK
- **AI:** Gemini + Groq (primary/fallback strategy)
- **Deployment:** Google Cloud Run

AI falls back to a keyword/rule engine if no API key is available — ensuring the system never goes dark.

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Live Demo Walkthrough

1

Open Dashboard

Show live zones, heatmap, and system health panel.

2

Register Volunteer

Start the Telegram bot, assign an ID and zone.

3

Send Normal Update

Submit a routine crowd status — watch the dashboard update live.

4

Trigger Emergency

Send "Gate A got fire" — observe auto-broadcast and AI suggestion.

5

Show Logs

Review system logs and the emergency state indicator in admin view.

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The Impact

<5s

Alert Broadcast

Emergency fan-out reaches all registered volunteers in under 5 seconds.

85%

Congestion Threshold

Automated alerts fire when zone density exceeds 85% capacity.

0

Manual Triage

AI classifies, prioritises, and routes every incident automatically — no dispatcher required.

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Ready to Deploy

One-liner

StadiumAI turns chaotic crowd updates into coordinated, AI-guided action in real time — keeping volunteers informed and attendees safer.

Next Steps

- Rotate API keys before public deployment
- Tighten Firestore security rules for production
- Move client-side AI calls server-side for key protection
- Add role-based access control to the admin dashboard
- Deploy to Google Cloud Run using included scripts

Dashboard live at: **stadiumai-1001460310030.us-central1.run.app**

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